# EDGE AI: What Hardware Designers Must Know

As edge AI chips surpass 900MHz, legacy hardware methods guarantee system failure. For designers, it's adapt or go obsolete.



**S.A. SRINIVASA MOORTHY** is the chief of strategy at Zettaone Technologies Private Limited. He is a prominent technology evangelist and subject matter expert in electronic design and manufacturing, automotive electronics, and electric vehicles.

With the hype around artificial intelligence (AI) reaching a crescendo, you might have noticed a curious silence on social media. Unlike software, there is no loud noise or marketing campaigns peddling courses like "*If you don't learn this, you will become obsolete*" for embedded systems.

Do not be fooled by the silence. Heed it as a warning before a massive industry disruption, the lull before the storm.

The first signs are already visible. Major semiconductor vendors are aggressively promoting AI-ready processors for edge processing. While many assumed that high memory prices would delay embedded AI/ML adoption, vendors have bypassed this hurdle. They are launching edge processors that operate at 900MHz and above.

For the average hardware engineer, this changes everything. Operating at frequencies above 300MHz pushes digital design straight into the analogue domain. Unfortunately, our education system remains stuck teaching obsolete 5V TTL logic, completely ignoring high-speed design, printed circuit board (PCB) layouts for high-speed circuits, and on-board DC-DC converters built for sub-1V (as low as 0.7V) requirements.

To survive and thrive in the AI/ML era, current-generation hardware designers must master the following baseline disciplines:

## Holistic system understanding

Without understanding the overall system architecture and data flow, designing hardware using high-speed AI/ML processors is impossible.

## Solid foundations in high-speed design

Many designers mistakenly assume high-speed design is just a minor signal integrity issue. It is not. True high-speed design requires deep domain knowledge across multiple physical layers. Thanks to open source tools like KiCAD, teaching and learning high-speed design is no longer a resource bottleneck. Designers must master:

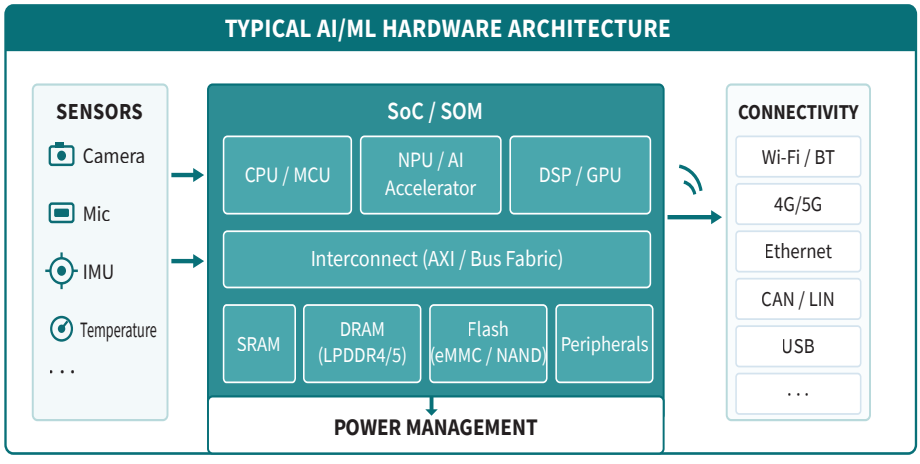
***PCB material selection.*** Choosing the right substrates to handle gigahertz frequencies.

***High-speed signal routing.*** Managing trace width, thickness, and routing paths to match electrical impedance, ensuring gigahertz signals flow smoothly without bouncing back or corrupting data.

***Power distribution networks.*** Designing stable on-board DC-DC power supply units for low-voltage cores.

***Thermal design.*** Mitigating severe heat dissipation from high-frequency chips.

***EMC/EMI design.*** Engineering layouts to pass strict electromagnetic compatibility and interference standards.




## Salient points

Embedded AI/ML systems are no longer isolated devices; they